

Ujaan Das

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EDUCATION

University of Warwick

Master of Science in Computer Science

Hong Kong University of Science and Technology

Bachelor of Engineering in Computer Engineering

Coventry, United Kingdom

Sep. 2025 – Sep. 2026

Hong Kong SAR

Sep. 2021 – May 2025

EXPERIENCE

Software Engineering Intern (Part Time)

Jan. 2025 – Aug. 2025

Stellerus Technology

Hong Kong SAR

- Re-designed a chatbot execution pipeline using a multi-step LLM workflow, implementing a custom command DSL for structured tool orchestration that reduced token overhead by 25% per session
- Standardized message contracts and schema enforcement across the pipeline, eliminating 90% of parsing-related regressions and reducing runtime errors by 30%
- Co-developed the SpaceGPT VLM platform awarded a Gold Medal at the 50th Geneva International Invention Exhibition; engineered the technical demo and automated deployment via Docker and Github Actions

Research Assistant

May 2024 – Sep. 2024

CASTLE Lab, HKUST

Hong Kong SAR

- Engineered high-throughput semantic-analysis pipelines in Python and Java to extract semantic signals from 30k+ scraped GitHub repositories to inform research in LLM-assisted testcase generation
- Integrated Tree-sitter to parse multi-million-line codebases into ASTs and CFGs, converting complex source code into lightweight, queryable intermediate representations
- Optimized this parsing logic to cut per-project analysis time by 45%, ultimately increasing usable data yield by 3.2x to support large-scale experiments, slated for publication in ACM ICSE'27

Software Engineering Intern

May 2023 – Aug. 2023

Stellerus Technology

Hong Kong SAR

- Developed RESTful APIs using FastAPI and MongoDB to orchestrate multi-user task queues for live satellite data, designed to sustain throughput of 8,000+ requests/sec with <150ms latency and minimal queue times
- Parallelized CPU-bottlenecked operations via a Celery/Redis worker pool, reducing processing time by 34%
- Partnered with frontend team to containerize the SPA and backend using Docker and orchestrated automated CI/CD pipelines for Azure Container Apps, cutting deployment time by 20%

PROJECTS

MSc Dissertation - Invariants | C++, Gtest, CMake, Python, Pybind11, Nix

[Github](#)

- Engineering a specification DSL using semantic invariants and refinement types to guide LLM decoding via verifiable runtime constraints
- Architecting a C++ compilation runtime with Python bindings for seamless integration into inference pipelines, enforcing schema compliance with targeted <30ms overhead

Snip | C++, Termios, Gtest, CMake, Nix

[Github](#)

- Architected an async C++ TUI framework and Vim-style modal editor with a thread-safe, non-blocking event pipeline and background worker pools for deferred execution
- Integrated native LSP tooling for real-time diagnostics and built hermetic, reproducible environments via Nix for cross-platform consistency

yumcha | Go, C, CGo, Cocoa, Nix

[Github](#)

- Engineered a macOS tiling window manager in Go and C using Cocoa/CoreGraphics, implementing an i3-style tree layout engine for dynamic, scriptable window management without disabling SIP

Undergraduate FYP - Follow-Me Robot | C++, Python, ROS2, Raspberry Pi, Docker

[Github](#)

- Developed a real-time UWB tracking system via custom ROS2 Python/C++ nodes, applying multi-rate Kalman and EMA filtering for stable sub-centimeter localization under noisy conditions

SKILLS

Languages: C++, Python, Go, C, TypeScript, SQL, JavaScript, HTML/CSS

Frameworks & Libraries: GTest, CMake, FastAPI, LangGraph, Pydantic, Celery

Databases: MongoDB, PostgreSQL, MySQL, SQLite, Redis

Tools & Cloud: Docker, Nix, Azure, AWS, GitHub Actions, NGINX, Make, Git